SCI-AST v0.1

Scientist-facing rationale for astrometric coordinates

Stage B targeted implementation-blind scientific author draft
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Revision r0.3 draft, 2026-08-22

Status boundary. This concise narrative explains the canonical authority carried in the companion Engineering Conformance Specification. It is a Stage B targeted draft, not scientific approval. Implementation conformity, empirical adequacy, validation, freeze, observational performance, and production readiness remain unassessed.

How to read this rationale

SCI-AST answers a narrow scientific question: for a particular detector occurrence and a particular aligned reference-grid slot, where is the detector looking, and which later coordinate facts can be derived from that direction? The answer is not merely two floating-point numbers. It includes the observation, detector occurrence, stable ALIGN slot, selected geometry, pointing correction, observing state, frame, projection, validity, uncertainty availability, and provenance that make the coordinate interpretable.

The authoritative import is exactly `SCI-ALIGN_TO_SCI-ASTv0.1/r0.1`. Compatibility requires preservation of all required identities, semantic states, ownership boundaries, and typed availability. A successor must name the revision it supersedes and map every changed, removed, or newly required item; similar fields or array shape do not establish compatibility.

Three notation choices prevent accidental aliasing. The stable ALIGN detector-reference slot is s, with stable identity (o, s). The symbol j is only a local storage row and is never an external identity. The stable RTC output sample is n. Finally, x and r are reserved exclusively for the paired KID readout coordinates; a unit sky direction is $\widehat{\boldsymbol{u}}_{\mathrm{sky}}$.

Narrative map: coordinate meaning → immutable parent → ordered geometry → frames/TAN → WCS/pixels → RTC response → MAP handoff.

1 What an AST coordinate scientifically means

An AST coordinate is a role-factored scientific record. A detector sky direction has the parent

$$\Pi_{\text{direction}} = (o, d, s,\ \text{exact ALIGN relation},\ \text{observing state},\ \text{pointing correction},\ \text{geometry/rotation realization}, F).$$

The tangent role adds the upstream-selected physical center and tangent projection. The continuous-pixel role adds the exact WCS. The nominal-pixel role then adds the rounding rule, dimensions, and bounds convention. These layers matter: WCS failure cannot erase a valid direction or tangent fact, and a nominal-pixel failure cannot erase a valid continuous pixel. A product family may demand a complete atomic product and fail it, while the available upstream scientific facts and exact causes remain intact.

Each role carries its own validity, parent, uncertainty availability, and four-stage provenance. Continuous WCS coordinates, nominal containing pixels, zero-based local indices, in-bounds state, MAP support, and coordinate precision are distinct facts.

2 The immutable ALIGN parent and signal validity

ALIGN supplies one immutable occurrence/time/mapping and observing-state relation. It also transfers the paired detector-readout source relation

$$x^A = A_{\text{ALIGN}} x^{\text{acq}}, \qquad r^A = A_{\text{ALIGN}} r^{\text{acq}}.$$

The two readout coordinates share native occurrences, weights, residuals, stable slots, source relation, and origin/synthesis state, while their numeric validity and payload availability remain separate. AST needs the exact occurrence, time, mapping, observing state, and geometry inputs; it does not need a finite or requested x^A or r^A payload to define an otherwise valid coordinate.

Five causes therefore remain independent: detector-occurrence identity, readout-coordinate validity, ALIGN mapping validity, AST coordinate validity, and downstream use-specific eligibility. No cause rescues, overwrites, or silently stands in for another.

3 Boresight, pointing correction, geometry, and field rotation

The operations are ordered and generally noncommuting. AST first admits the aligned boresight and current science-occurrence state. It applies the producer-selected pointing correction only inside its selected native support, then evaluates the selected geometry and field rotation at the current aligned time and elevation. Current elevation is never inferred from a bracketing pointing record. AST neither extrapolates nor selects a nearest or stale correction.

For tangent components $\boldsymbol{\eta} = (\eta_1, \eta_2)$ in the local orthonormal basis $(\boldsymbol{e}_1, \boldsymbol{e}_2)$, define

$$\boldsymbol{t} = \eta_1 \boldsymbol{e}_1 + \eta_2 \boldsymbol{e}_2, \qquad \rho = \|\boldsymbol{t}\|, \qquad \widehat{\boldsymbol{v}} = \boldsymbol{t}/\rho \quad (\rho > 0).$$

The complete spherical oracle is

$$\mathrm{Exp}_{\widehat{\boldsymbol{u}}_{\mathrm{sky}}}(\boldsymbol{\eta}) = \cos\rho\, \widehat{\boldsymbol{u}}_{\mathrm{sky}} + \sin\rho\, \widehat{\boldsymbol{v}} \qquad (\rho > 0),$$

with $\widehat{\boldsymbol{u}}_{\mathrm{sky}}$ returned at $\rho = 0$. The established small-angle production relation is allowed only within a preregistered adequate domain; the spherical relation is the independent offline oracle, not an automatic runtime fallback.

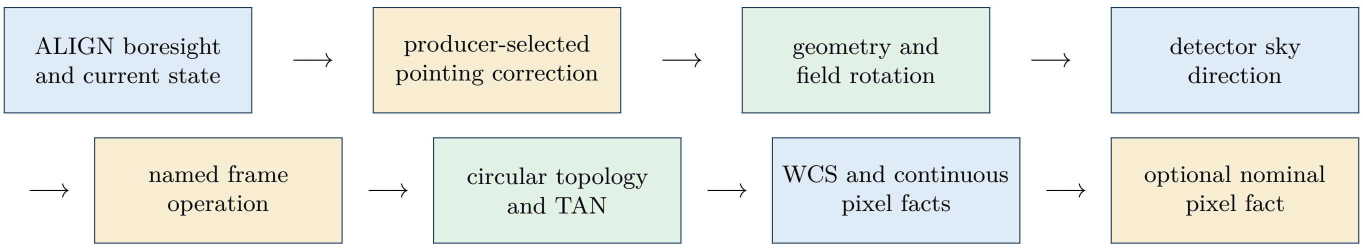

Figure 1: The ordered, noncommuting AST chain. Every stage retains its exact parent and application count; a special zero-offset case does not authorize reordering or double application.

4 APT gauge, pivot uncertainty, affine limitations, and transfer

SCI-BEAM owns measured relative detector geometry and its limitations. An exact named TolProj/TolAPT or other approved authority owns the observation-specific association/transformation and immutable APT realization. The selected artifact declares whether its detector coordinates are raw observation-local, horizon-fixed/derotated, or another exact named representation. It declares coordinate basis and handedness, angular units, origin/recentering gauge, every field-rotation or derotation operation already embodied and its application count, field-rotation law and support, conventional pivot state, covariance availability, unresolved translation/scale/skew/shear, and complete lineage. AST consumes this artifact; it does not infer that the APT origin is telescope boresight, nominate a reference detector, or silently reduce the relation to a pure rotation about a known pivot. AST applies only a transformation not already embodied by that declaration. It never double-derotates or infers representation from a field name or numerical value.

Pointing-support observations and the corrected science observation must share the same realization key

$$K_\gamma = (\text{APT realization},\ \text{gauge},\ \text{geometry version},\ \text{rotation law},\ \text{pivot},\ \text{AST convention}).$$

A mismatch is incompatible unless a separately authorized transform names both immutable parents, proves equivalence over a declared domain, acts in the correct direction and units, and propagates covariance, cross-covariance, and transform uncertainty.

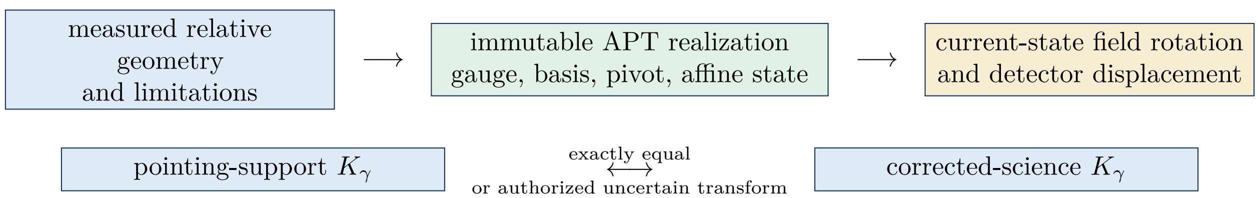

Figure 2: Gauge and field rotation remain part of one immutable realization. Pointing transfer requires the same realization or a complete, explicitly uncertain equivalence transform.

5 Frame conversion, circular topology, and TAN

Point, OOF, and Beammap retain native AltAz tangent coordinates in arcseconds. Science uses explicitly identified equatorial J2000 TAN coordinates in degrees. J2000, FK5, ICRS, apparent place, observed AltAz, and unrefracted AltAz are not synonyms. Every non-identity transform declares source and target frames, epoch/equinox, time scale, site, model/version, EOP/refraction applicability, inputs, validity, and uncertainty availability.

For period P, shortest signed difference is exactly

$$\mathrm{wrap}_P(\theta) = \theta - P \left\lfloor \frac{\theta + P/2}{P} \right\rfloor \in [-P/2, P/2).$$

Persisted longitude lies in $[0, P)$. Exactly antipodal interpolation has no unique shortest path and is unavailable unless an explicit unwrap authority selects and parents the path. At a coordinate pole the unit vector, not an ambiguous longitude, is the direction identity.

For direction $\widehat{\boldsymbol{u}}_{\mathrm{sky}}$, selected center $\widehat{\boldsymbol{u}}_{\mathrm{sky},0}$, and center basis $(\boldsymbol{e}_1, \boldsymbol{e}_2)$, TAN uses

$$D = \widehat{\boldsymbol{u}}_{\mathrm{sky}} \cdot \widehat{\boldsymbol{u}}_{\mathrm{sky},0}, \qquad \zeta_a = \frac{\widehat{\boldsymbol{u}}_{\mathrm{sky}} \cdot \boldsymbol{e}_a}{D}, \quad a \in \{1, 2\}, \quad D > 0.$$

There is no universal positive $D_{\min}$. Non-finite input/output or $D \leq 0$ is coordinate invalidity, never a center, edge, or stale pixel.

Why the vector form matters

The unit-vector formulation remains finite through longitude wrap and near a pole, exposes the exact spherical domain, and prevents either paired readout coordinate x or r from being confused with sky direction or a TAN axis.

6 Physical center, WCS, continuous pixels, and nominal pixels

The physical center is selected by a named upstream target/product authority. AST realizes that exact choice; it never searches, fits, silently recenters, derives a center from bounds, or supplies a default. The exact six-field legacy all-zero control pattern adapts only at the boundary to `automatic`. A nonzero or non-finite legacy explicit request is unsupported in v0.1, while numeric zero elsewhere remains an ordinary value.

An immutable WCS binds projection, center authority, CRVAL, one-based CRPIX, CD or equivalent PC+CDELT, axes/sign/rotation/handedness, dimensions, bounds, units, frame/epoch, precision, product plan, and realization version. If $\boldsymbol{w}$ is the tangent coordinate and $\boldsymbol{q}$ the one-based continuous FITS coordinate,

$$\boldsymbol{w} = \mathrm{M}(\boldsymbol{q} - \boldsymbol{q}_0), \qquad \boldsymbol{q} = \boldsymbol{q}_0 + \mathrm{M}^{-1}\boldsymbol{w}.$$

When the product plan selects geometric array center, $\mathrm{CRPIX}_a = (N_a + 1)/2$, so even dimensions have half-integer reference pixels.

The v0.1 nominal containing pixel uses a half-open nearest-center rule. Local storage row j and column h are zero-based implementation-local positions, not external identities. In particular, no array reorder may change stable slot (o, s). A subpixel dither changes the continuous pixel smoothly; the nominal pixel changes only at its declared boundary. Neither fact establishes MAP coverage or coordinate precision.

Layered availability example. If the frame transform succeeds but the requested WCS is unavailable, the detector direction remains valid. If the selected center and TAN succeed but a pixel matrix is singular, the tangent coordinate remains valid. A family that requires a complete map coordinate product still fails that product; the upstream facts are not retroactively deleted.

7 ALIGN-grid versus RTC-output-grid coordinates

θ^{A}_{ds} belongs to the exact ALIGN grid and stable slot (o, s). $\theta^{\mathrm{RTC}}_{dn}$ belongs to the exact RTC product, resolved plan, output grid, stable output sample n, representative ALIGN slot, selected output time, phase/delay convention, segment, decimation, support, response, and correction state. Even if a phase-zero RTC coordinate is numerically equal to one ALIGN-grid coordinate, their roles and parents differ.

For every ordinary downstream science product whose numerical signal is represented on an RTC output grid, AST realizes the corresponding stable role `SCI-AST:rtc_output_grid_coordinates@1`. RTC supplies the exact product and plan, grid and stable sample n, representative ALIGN slot, selected output time, phase/delay/time-shift convention, segment, decimation factor/phase/support, coordinate-correction state, temporal response, and full support. AST binds those facts without reconstruction and owns the detector direction plus requested tangent/WCS/pixel roles, coordinate validity, uncertainty availability, and provenance on that exact grid. Missing RTC parent facts make only this role unavailable.

For RTC signal operator L_{ns}, role-factored sky response retains the operator and the full contributing coordinate support:

$$\text{sky-response}(L_{ns}, \{\theta^{A}_{ds}\}) \neq \sum_{s} L_{ns}\theta^{A}_{ds}.$$

Angular coordinates are not detector signals. Negative coefficients, circular wrap, and nonlinear projection make direct filtering of angles an incorrect replacement for the response support.

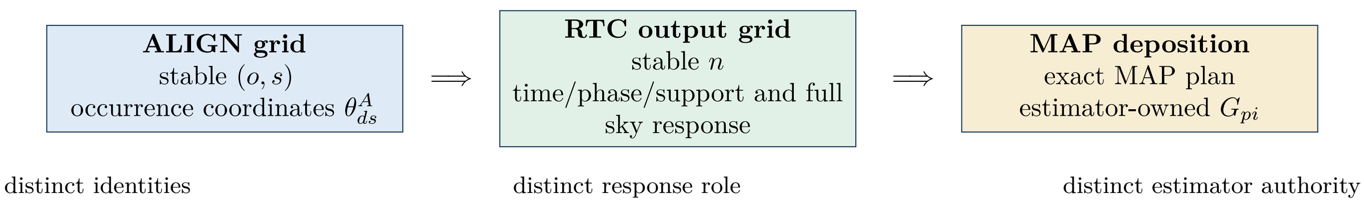

Figure 3: ALIGN-grid coordinates, RTC-output-grid coordinates, and MAP deposition are three distinct scientific stages. Numerical coincidence or equal shape never aliases their parents.

8 AST coordinates versus MAP G_{pi}

AST owns the occurrence-to-direction, tangent, WCS, and continuous-pixel geometry. MAP owns how a sample contributes to pixels: kernel family and parameters, estimator support, normalization, edge behavior, conservation, response, covariance, accumulation, cropping, and reprojection.

The symbol G_{pi} is reserved for an exact MAP-owned deposition/gridding plan. AST may materialize it only as a delegated service for a complete exact MAP request, with the AST coordinate/WCS parent, applicable RTC parent, and MAP plan all bound. Delegation transfers no scientific ownership:

AST determines exactly where the occurrence is; MAP determines exactly how it contributes to pixels.

Before a complete MAP request, the base AST relation ends at continuous pixel, optional nominal containing pixel under a declared rounding rule, and bounds state. It publishes no general neighbor stencil or candidate support whose extent depends on a MAP kernel, radius, support, edge rule, normalization, or conservation behavior. Any such relation requires the exact MAP-owned plan. Conversely, two MAP plans can use the same AST continuous coordinates to produce different valid G_{pi} operators without changing the AST coordinate identity.

Ordinary nonpolarimetric coordinate path only

Only the ordinary nonpolarimetric coordinate path is in scope. AST does not interpret polarization. Optional HWPR timing may remain only as an ALIGN-parent fact; it authorizes no demodulation, polarization calibration or response, or Stokes reconstruction. Raw KID readout x is not identified as Stokes I.

The AST–MAP ownership test

Within this boundary, a choice belongs to MAP when it changes the sample-to-pixel deposition kernel, normalization, boundary treatment, conservation rule, or realized numerical map contribution without changing the AST coordinate. Upstream analysis coefficients remain owned by PTC or their named producer; eligibility remains owned by the named-use profile and evaluator; calibration, response, and uncertainty retain their existing owners. A choice affecting estimator weight is therefore not automatically a MAP choice.

9 Astrometric uncertainty, validation, and open decisions

The only v0.1 astrometric response product is the typed `astrometric_map_center_jacobian`. It exists only after a named family requests exact input/output domains, direction, axes, and units. It is evaluated at the declared center and is not an amplitude, beam, RTC temporal, MAP transfer, or point-source response. Only declared available covariance and cross-covariance terms are propagated; unavailable is not zero, and no total is formed without complete combination assumptions.

Open owner decisions

Seven existing questions remain open or deferred: aligned-field registry and composition history; per-family center selection; covariance producers and conditioning; quantitative small-angle/time/precision adequacy; MAP-003 retained-grid disposition; geometry association and cross-realization transform; and family-specific Jacobian requests. AST-OWNER-Q008 is closed for the ordinary science chain by the RTC-grid role above. Each remaining question blocks only its dependent output or claim. This targeted revision introduces no new owner question.

Compact rationale-to-contract crosswalk

Narrative subject	Engineering authority
Immutable ALIGN parent and cause separation	REQ-006–012; PRED-009–010
Correction, geometry, gauge, composition	REQ-013–035; PRED-001–017
Direction, frame, topology, TAN	REQ-036–044; PRED-018–022
Center, layered WCS and pixel facts	REQ-045–064; PRED-023–031, 041–044
Jacobian and covariance	REQ-065–072; PRED-032–034
ALIGN/RTC/MAP role boundaries	REQ-073–083; PRED-035–040
Provenance, adequacy, claim separation	REQ-084–090; PRED-045–050

The Engineering Conformance Specification retains the complete notation, definitions, canonical equations, 15 assumptions, 90 stable requirements, and 50 stable predictions. Those predictions are future falsifiers, not reported test results. One clean horizontal ALIGN–AST audit remains required before any scientific freeze. This document's deterministic build and visual review are publication checks only and establish none of the unassessed claims named on the first page.